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Dual-Axis Solar Tracker

Project Report

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Abstract

Solar energy harvesting efficiency is critically dependent on panel-sun alignment, yet most small-scale PV installations use fixed mounts that operate at sub-optimal angles for most of the day, capturing only 60–70% of potential energy. The primary challenge is that conventional tracking systems are either too expensive or too complex for small-scale educational and research applications, limiting their adoption in resource-constrained environments.

This project develops a low-cost dual-axis solar tracker that continuously aligns a 5–15 W photovoltaic panel perpendicular to the sun’s rays to maximize energy harvest. The system employs orthogonal azimuth and elevation axes driven by servo motors, controlled by an Arduino microcontroller using a hybrid approach combining LDR sensor feedback and sun-position algorithms.

Contents

1 Introduction	3
1.1 Objectives	3
2 Background and Literature Review	3
2.1 Solar Energy and the Need for Tracking Systems	3
2.2 Classification of Solar Tracking Systems	3
2.3 Control Strategies for Solar Tracking	4
2.3.1 Sensor-Based Control	4
2.3.2 Algorithmic Control	4
2.3.3 Hybrid Control	5
2.4 Mechanical Design Considerations	5
2.5 Actuator Technologies	5
2.6 Power Budget and Energy Efficiency	6
2.7 Control System Architecture	6
2.8 Performance Metrics and Validation	6
2.9 Relevance to Current Project	6
3 Alternative Designs and Selection Process	7
3.1 Proposed Design Alternatives	7
3.1.1 Design A: Passive Seasonally-Adjusted Fixed Mount	7
3.1.2 Design B: Single-Axis Azimuth Tracker	7
3.1.3 Design C: Dual-Axis Active Tracker (Selected Design)	8
3.2 Selection Criteria and Evaluation	8
3.3 Selection Justification	9
4 System Overview	10
5 Mechanical Design	11
5.1 Frame and Mounting	11
5.2 Actuator Selection and Structural Considerations	11
6 Electrical Design	11
6.1 Controller and I/O	11

7 Control Algorithm Approach	12
7.1 LDR Differential (Sensor) Approach	12
8 Bill of Materials (BOM)	12
9 Manufacturing and Assembly	13
9.1 Mechanical Parts	13
9.2 Electrical Assembly	13
10 Completed design and assembly	13
11 Conclusions and Future Work	13

1 Introduction

Solar energy capture is highly sensitive to the angle of incidence. Fixed panels are subject to sub-optimal exposure throughout the day and seasons; a dual-axis tracker can increase energy capture substantially by keeping the panel normal to incident sunlight. This report describes the design and implementation strategy of a compact, low-power dual-axis tracker suitable for small panels (5–15 W class) for research and educational prototyping.

1.1 Objectives

- Design a rigid mechanical frame supporting two orthogonal axes (azimuth and elevation).
- Implement a control system (microcontroller + sensors) to track the sun continuously.
- Ensure the unit is energy-positive: tracker energy consumption $\ll$ incremental energy harvested.
- Produce CAD models and a bill of materials (BOM) for manufacturing and assembly.
- Test and validate tracking accuracy and energy yield improvement over a fixed mount.

2 Background and Literature Review

2.1 Solar Energy and the Need for Tracking Systems

Photovoltaic (PV) panel efficiency is highly dependent on the angle of incidence between incoming solar radiation and the panel surface. When sunlight strikes a solar panel at oblique angles, energy absorption is significantly reduced due to increased reflection and reduced effective surface area. Fixed-mount PV installations experience substantial variations in energy capture throughout the day as the sun traverses the sky, with maximum power output achieved only during brief periods when the sun is near-optimal alignment with the panel [1].

Studies have demonstrated that dual-axis tracking systems can increase energy yield by 30–45% compared to fixed-tilt installations, with exact improvements depending on geographic location, seasonal variations, and local weather patterns [2, 3]. This substantial gain in energy capture makes solar tracking particularly attractive for small-scale and research applications where maximizing output from limited panel area is critical.

2.2 Classification of Solar Tracking Systems

Solar tracking systems are broadly categorized into two classes based on degrees of freedom: single-axis and dual-axis trackers. Single-axis systems rotate about one axis (typically azimuth for horizontal trackers or tilt for vertical trackers) and offer a compromise between complexity and performance gain, achieving 20–30% improvement over fixed systems [4]. Dual-axis trackers, by contrast, employ two orthogonal axes of rotation—azimuth (horizontal rotation) and elevation/altitude (vertical tilt)—enabling the

panel to maintain near-perpendicular alignment with the sun throughout the day and across seasons. This configuration yields the highest possible energy capture but introduces additional mechanical complexity and control requirements.

2.3 Control Strategies for Solar Tracking

The literature identifies three primary control strategies for solar tracking systems: sensor-based, algorithmic (or open-loop), and hybrid approaches.

2.3.1 Sensor-Based Control

Sensor-based systems utilize photodetectors—commonly light-dependent resistors (LDRs), photodiodes, or solar cells—arranged in differential configurations to detect the direction of maximum light intensity. The most common configuration employs four sensors in a quadrant arrangement with physical shading fins or barriers between them. The control algorithm compares sensor readings and actuates motors to minimize differential error, thereby aligning the panel normal to the sun [5, 6].

Advantages:

- Simple implementation with minimal computational requirements
- Self-correcting behavior compensates for mechanical inaccuracies
- Adapts automatically to local atmospheric conditions and diffuse light

Disadvantages:

- Poor performance in cloudy or low-light conditions
- Susceptible to false triggers from reflections or nearby artificial light sources
- Requires periodic cleaning and maintenance of sensors
- May experience hunting behavior or oscillation without proper deadband implementation

2.3.2 Algorithmic Control

Algorithmic tracking systems compute solar position angles (azimuth and elevation) based on date, time, and geographic coordinates using well-established solar geometry models. Common algorithms include the simplified Solar Position Algorithm (SPA) developed by NREL [7] and various analytical approximations suitable for microcontroller implementation. These systems use real-time clock (RTC) modules and, optionally, GPS for precise location and time synchronization.

Advantages:

- Consistent performance independent of weather conditions
- Predictable behavior enables energy-efficient scheduled movements
- No sensor maintenance or cleaning required
- Can incorporate sleep modes during nighttime to conserve energy

Disadvantages:

- Requires accurate initial configuration of location and time
- Accumulation of mechanical backlash or positioning errors over time
- No self-correction capability for mounting misalignment
- Higher computational overhead for embedded systems

2.3.3 Hybrid Control

Hybrid systems combine algorithmic coarse positioning with sensor-based fine-tuning, leveraging the strengths of both approaches [8, 9]. The algorithm provides initial positioning, reducing sensor hunting time and enabling operation during marginal light conditions, while the sensor array corrects for accumulated positioning errors and mechanical tolerances.

2.4 Mechanical Design Considerations

The mechanical subsystem of a solar tracker must balance structural rigidity, actuation torque requirements, and cost. Key design parameters include:

- **Material Selection:** Aluminium extrusions and steel components are common for high-rigidity applications, while wood or PVC may be suitable for lightweight prototypes and educational demonstrations. Material choice directly impacts system mass, which in turn influences actuator sizing [10].
- **Bearing Selection:** Rotational axes require appropriate bearing types to minimize friction and prevent binding. Ball bearings or sleeve bearings are typical for low-to-moderate load applications. Proper bearing selection ensures smooth motion and reduces actuator power consumption.
- **Torque Requirements:** Actuator torque must account for panel weight, center-of-gravity offset, wind loading, and friction losses. Wind load is a critical consideration for outdoor installations; drag forces scale with panel area and can generate significant torques at even moderate wind speeds [11]. A common design practice applies a safety factor of 1.5–2.0 to calculated torque requirements.
- **Limit Switches and Safety:** Mechanical stops and electrical limit switches prevent over-rotation, protecting both the structure and wiring from damage. Emergency shutoff provisions are essential for systems exposed to high wind conditions.

2.5 Actuator Technologies

Several actuator technologies are employed in solar tracking applications:

- **DC Geared Motors:** Provide continuous rotation with high torque output. Require H-bridge motor driver circuits for bidirectional control. Simple to interface with microcontrollers but lack inherent position feedback.
- **Stepper Motors:** Offer precise angular positioning in discrete steps without requiring encoders. Well-suited for open-loop algorithmic control but may draw significant holding current.
- **Servo Motors:** Integrate position feedback and control electronics, simplifying system design. Continuous rotation servos or standard positional servos may be used depending on range requirements. Limited to specific angular ranges unless modified.

- **Linear Actuators:** Provide push-pull motion for tilt axis control, converting linear motion to angular displacement. Often used in larger-scale trackers due to high force capacity.

For small-scale prototypes (5–15 W panel class), servo motors or small DC geared motors typically provide adequate performance with minimal power consumption [12].

2.6 Power Budget and Energy Efficiency

A design constraint for solar trackers is ensuring that the tracking system remains energy-positive: the incremental energy gained through tracking must significantly exceed the energy consumed by the control system and actuators [13]. Key power management strategies include:

- Intermittent tracking: moving in discrete steps rather than continuous adjustment
- Sleep modes for microcontrollers during idle periods
- High-efficiency motor drivers and voltage regulators
- Minimizing mechanical friction through proper bearing selection and lubrication

2.7 Control System Architecture

Modern solar trackers typically employ low-cost microcontroller platforms such as Arduino, ESP32, or similar embedded systems. These platforms provide:

- Analog-to-digital conversion for sensor interfacing
- PWM outputs for motor control
- Serial communication for debugging and data logging
- Sufficient computational capacity for real-time control algorithms
- Low power consumption suitable for PV-powered operation

Arduino-based implementations are particularly popular in educational and research settings due to extensive community support, readily available libraries, and low barrier to entry [14].

2.8 Performance Metrics and Validation

Solar tracker performance is evaluated using several key metrics:

- **Tracking Accuracy:** Angular error between panel normal and sun vector, typically measured in degrees. High-performance systems achieve ± 0.1 accuracy, while low-cost educational systems may operate at ± 2 – 5 .
- **Energy Gain:** Percentage increase in daily or annual energy harvest compared to optimally-tilted fixed installation. This is the primary performance indicator.
- **Power Consumption:** Average and peak power draw of tracking system, typically compared against panel output.
- **Response Time:** Speed at which the tracker responds to changes in solar position or sensor input.

2.9 Relevance to Current Project

This project develops a compact dual-axis solar tracker suitable for small PV panels (5–15 W class). Based on the reviewed literature, the design adopts:

- A hybrid control approach utilizing both LDR sensor arrays and optional algorithmic positioning for robustness
- Lightweight mechanical construction (wood-based frame) to minimize actuator torque requirements and reduce cost
- Arduino-based microcontroller platform for accessibility and ease of programming
- Servo motors for actuation, balancing cost and performance

3 Alternative Designs and Selection Process

3.1 Proposed Design Alternatives

To address the solar tracking challenge for small-scale PV applications, three distinct design alternatives were developed and evaluated. Each alternative represents a different approach to the fundamental trade-off between system complexity, cost, and energy capture performance.

3.1.1 Design A: Passive Seasonally-Adjusted Fixed Mount

This design employs a manually adjustable tilt mechanism that allows the user to set the panel angle at the beginning of each season (quarterly adjustments). The mount consists of:

- Fixed base platform with adjustable angle brackets
- Manual locking mechanism with discrete tilt positions (15°, 30°, 45°, 60°)
- North-facing orientation (for Southern Hemisphere installations)
- No electronic components or power consumption

Strengths:

- Minimal construction complexity; requires only basic carpentry skills
- Lowest cost
- Zero power consumption and maintenance requirements
- High reliability due to absence of moving parts and electronics
- Suitable for very remote locations without access to replacement parts

Limitations:

- Limited energy gain (5–10% over fixed horizontal mount)
- Requires manual intervention four times per year
- No intra-day tracking of sun movement
- Minimal educational value for mechatronics learning
- Cannot respond to local weather or shading conditions

3.1.2 Design B: Single-Axis Azimuth Tracker

This design implements automated east-to-west tracking through horizontal rotation while maintaining a fixed tilt angle optimized for local latitude. The system includes:

- Rotating platform mounted on vertical axis with thrust bearing
- Single servo motor or geared DC motor for azimuth control
- Dual LDR sensors (east-west pair) for sun position detection
- Arduino Nano microcontroller and motor driver
- Fixed elevation angle set to latitude + 15° (seasonal average)

Strengths:

- Moderate complexity; single-axis simplifies mechanical design and programming
- Significant energy gain over fixed mount
- Lower cost compared to dual-axis systems
- Reduced maintenance compared to dual-axis (fewer moving parts)
- Good educational value for demonstrating basic control concepts
- Lower wind load exposure compared to dual-axis configurations

Limitations:

- Sub-optimal performance during seasons when sun elevation differs from fixed angle
- Cannot compensate for latitude-specific seasonal variations
- Misses 10–15% energy capture potential compared to dual-axis
- Less comprehensive as a mechatronics learning platform
- Fixed tilt may require seasonal manual adjustment for optimal performance

3.1.3 Design C: Dual-Axis Active Tracker (Selected Design)

This design implements full two-axis control with independent azimuth and elevation tracking, enabling the panel to maintain near-perpendicular alignment with the sun throughout the day and year. The system comprises:

- Azimuth rotation platform with vertical axis servo
- Elevation tilt frame with horizontal axis servo
- Quadrant LDR sensor array (four sensors) with shading cross
- Arduino Uno R3 microcontroller with dual motor control
- Optional hybrid control: RTC module for algorithmic positioning + sensor fine-tuning
- Limit switches on both axes for safety

Strengths:

- Maximum energy capture potential (30–45% over fixed mount)
- Optimal alignment maintained throughout day and year
- Excellent educational platform demonstrating multi-axis control, sensor fusion, and embedded programming
- Scalable design applicable to larger systems
- Hybrid control provides robustness to varying weather conditions
- Demonstrates industry-relevant mechatronics integration

Limitations:

- Highest complexity; requires precise mechanical alignment and programming
- Moderate cost (3,000 KSh) due to additional motor and sensors
- Higher maintenance requirements (two motor systems, sensor cleaning)
- Increased power consumption compared to single-axis
- Greater wind load exposure requires stronger structural design

3.2 Selection Criteria and Evaluation

The selection of an optimal design requires balancing multiple competing objectives. A weighted decision matrix was developed using criteria relevant to the project’s educational and technical goals.

Table 1: Design comparison and selection matrix

Criterion	Weight (%)	Design A (Score)	Design B (Score)	Design C (Score)
Energy gain over fixed mount	40	3/10 (5–10%)	7/10 (20–30%)	10/10 (30–45%)
Construction complexity	15	10/10	7/10	5/10
Total system cost	20	10/10	7/10	5/10
Maintenance requirements	10	10/10	8/10	6/10
Weighted Score		6.35/10	7.20/10	7.55/10
Ranking		3rd	2nd	1st

Scoring methodology:

- Energy gain: Scored based on literature-reported performance improvements
- Complexity: Assessed based on number of axes, components, and fabrication steps
- Cost: Based on component BOM and material estimates
- Maintenance: Evaluated based on number of moving parts and sensor exposure

3.3 Selection Justification

Design C (Dual-Axis Active Tracker) was selected as the optimal solution despite its higher complexity and cost. The rationale for the selection is as follows.

Primary Justification : Maximum Energy Capture: The fundamental purpose of a solar tracker is to maximize energy harvest. Design C achieves 30–45% improvement over fixed mounts, representing a 10–15 percentage point advantage over the single-axis alternative (Design B) and a 20–35 percentage point advantage over the passive design (Design A). For small panel installations where every watt counts, this performance differential justifies the additional complexity.

Educational and Research Value:

- Multi-axis mechanical design and kinematic analysis
- Dual motor control and PWM signal generation
- Sensor array design and differential signal processing
- System integration and debugging of complex embedded systems

Scalability and Industry Relevance: The dual-axis control architecture is directly scalable to larger commercial tracking systems.

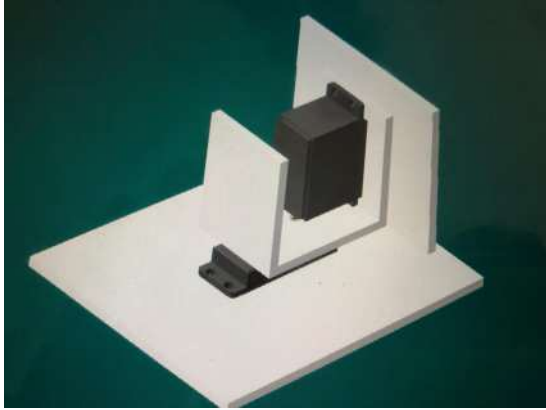
Wood-Based Construction Mitigates Complexity: The decision to use wood as the primary structural material significantly reduces the complexity concerns associated with Design C. Wood enables:

- Rapid prototyping with basic hand tools (no machining or welding required)
- Easy modifications and adjustments during development
- Lightweight construction reducing actuator torque requirements

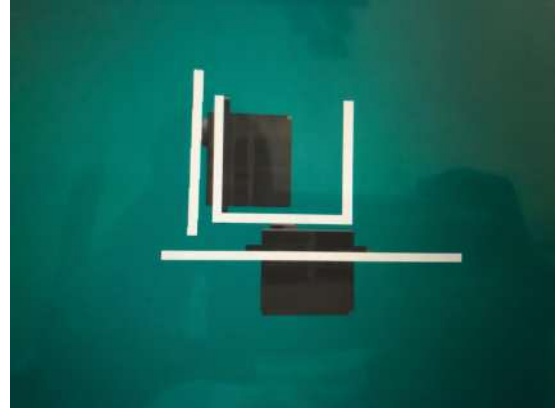
- Low-cost mistake tolerance (material can be easily replaced)

This material choice brings the practical complexity of Design C closer to Design B while retaining the full performance advantages.

4 System Overview

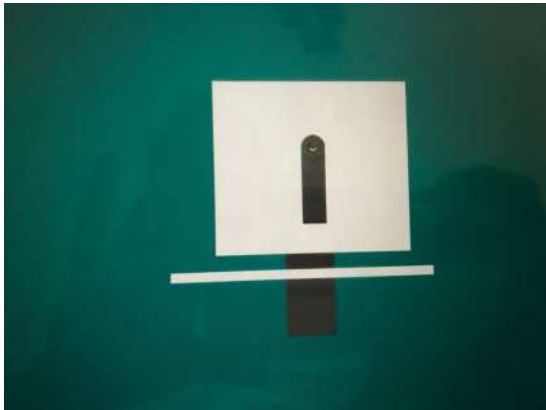


(a) Isometric view of tracker assembly

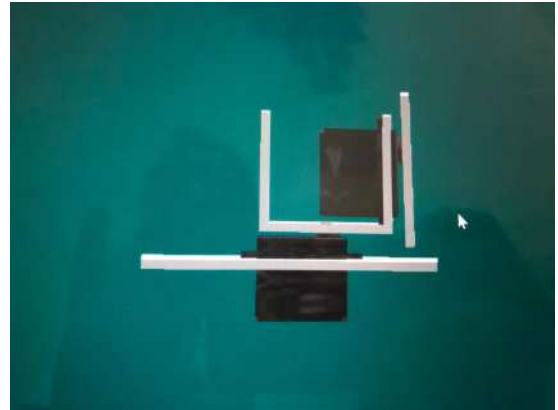


(b) Side elevation view

Figure 1: CAD models of dual-axis solar tracker mechanical design.



(a) Top view showing azimuth rotation



(b) Detail view of servo mounting

Figure 2: Additional CAD views of tracker components.

The system consists of:

- Mechanical subsystem: panel frame, azimuth axis, elevation axis, bearings, mounting base.
- Actuation: two geared motors (or stepper motors) with appropriate torque rating.
- Sensing: LDR sensor array (quadrant) with shading fins or a GPS/RTC-based algorithmic sun angle computation.
- Control: Microcontroller (Arduino Uno R3 clone in current BOM), motor drivers, limit switches/encoders.

- Power: small PV panel + battery (or using the panel under test for energy); power management circuitry to feed controller and actuators.

5 Mechanical Design

5.1 Frame and Mounting

The structural frame of the dual-axis solar tracker is fabricated primarily from wood, selected for its low cost, ease of machining, and local availability. Wooden members form the base platform, azimuth support, and elevation U-frame that holds the photovoltaic panel.

Key design features include:

- Plywood used for the base and vertical supports.
- Screwed and glued joints to improve stiffness and reduce loosening under vibration.
- Wooden U-frame supporting the panel about the elevation axis.

Although wood is more susceptible to environmental effects than metal, the tracker is intended primarily for indoor demonstration or short-term outdoor testing. Protective coatings such as varnish or paint may be applied to improve durability.

5.2 Actuator Selection and Structural Considerations

The use of wood as the primary structural material reduces overall system mass, lowering the torque required from the actuators. This allows the use of small servo motors or low-power geared DC motors for both azimuth and elevation control.

Key considerations include:

- Lower stiffness compared to metal, requiring careful joint design to minimize flexing.

6 Electrical Design

6.1 Controller and I/O

- **Microcontroller:** Arduino Uno R3 — reads sensors, computes control values, issues motor driver commands.
- **Sensors:** Light Dependent Resistor arranged under shading fins for sensor approach.

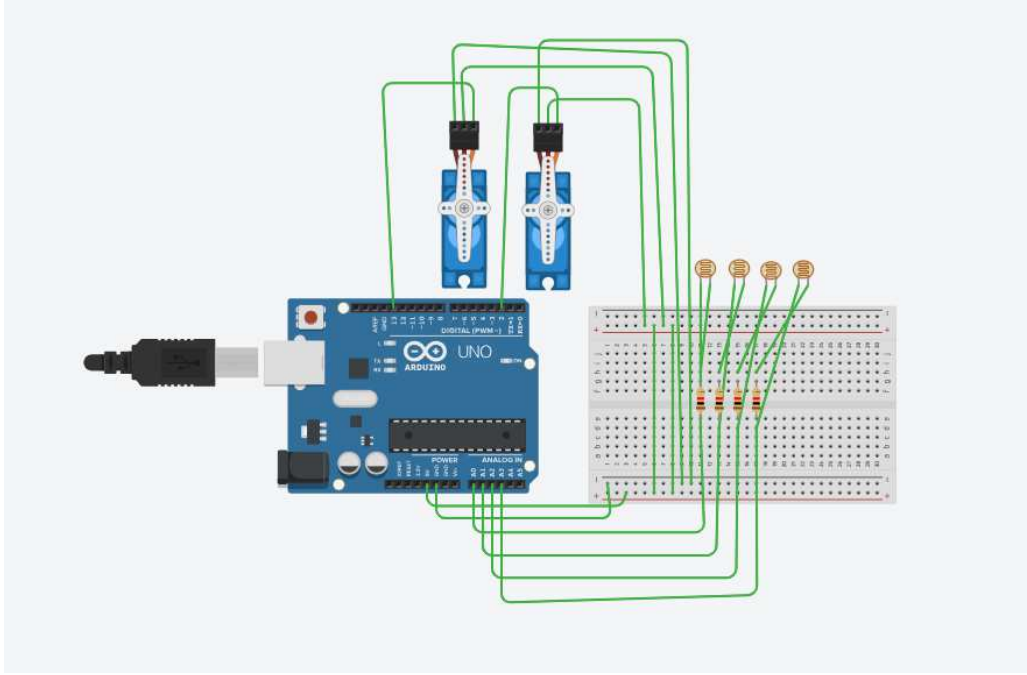


Figure 3: Circuit Schematic from TinkerCAD adapted from [15].

7 Control Algorithm Approach

7.1 LDR Differential (Sensor) Approach

1. Four LDRs form quadrants: compare left vs right for azimuth, top vs bottom for elevation.
2. If difference $>$ threshold, rotate axis toward higher intensity until balanced.

8 Bill of Materials (BOM)

Component	Quantity	Est. Unit Price (KSh)	Total (KSh)
Arduino Uno R3 (Clone)	1	1,500	1,500
Servo Motor SG90 (blue, plastic)	2	350	700
LDR Sensors (5 mm)	4	50	200
Solar Panel (Small 5V/6V)	1	400 – 500	450
Breadboard (Small/Medium)	1	200 – 300	250
Jumper Wires (male-to-male, 1 pkt)	1	150 – 200	150
Resistors (10k Ω)	4	5	20
Plywood / Wooden Boards (12–18 mm)	As required	—	—
Wood Glue / Adhesive	1 tube	—	—
Estimated TOTAL			3,270 KSh

Note: Wooden structural components were sourced locally and therefore not included in the electronic cost total.

9 Manufacturing and Assembly

9.1 Mechanical Parts

The mechanical components are manufactured using basic woodworking tools, making the design suitable for low-resource environments.

Manufacturing steps include:

- Cutting wooden planks to required dimensions using hand saws or power saws.
- Assembling the frame using wood screws, wood glue, and metal brackets where necessary.
- Sanding edges and applying protective finishes (paint or varnish).

9.2 Electrical Assembly

- Assemble breadboard prototype first to test control code.

10 Completed design and assembly

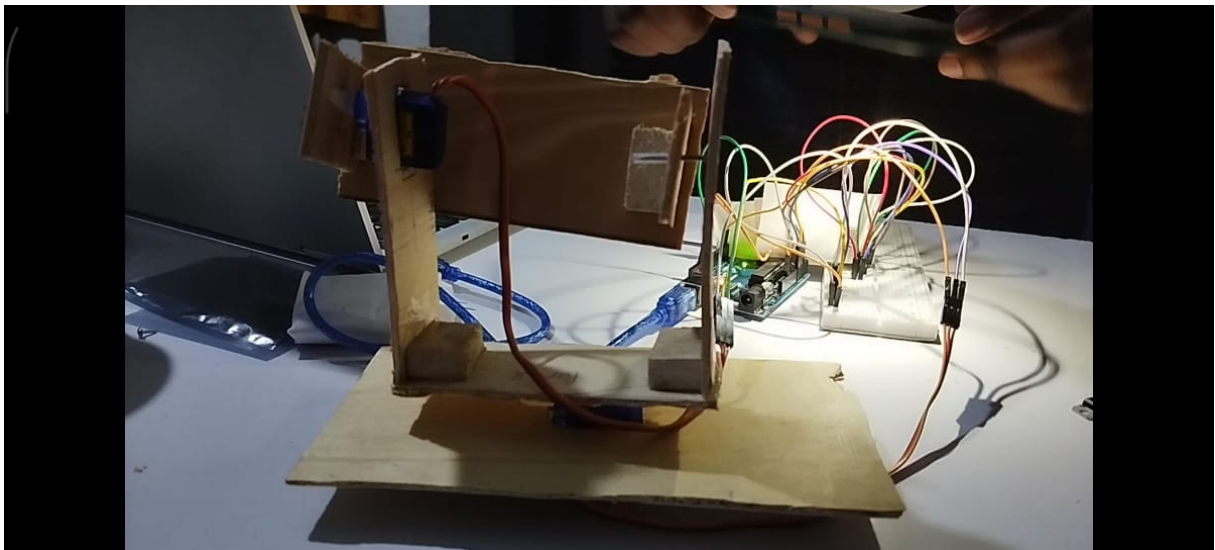


Figure 4: Completed Assembly of the Solar Tracker and in motion.

11 Conclusions and Future Work

This design provides a practical, low-cost dual-axis tracker that can be built with locally available components. The use of wood as the primary structural material significantly reduced manufacturing cost and complexity, making the tracker suitable for rapid prototyping and educational applications. Future improvements may include:

- Upgrading to closed-loop position feedback (encoders) for precise angular control.
- Implementing power-aware scheduling (limit movements under low energy).
- Scaling the mechanical design to larger panels with stronger actuators and structural reinforcement.

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